User Guide

Trajectory Modeling for One or Two Continuous Outcomes (Censored Normal)

File: `continuous_2outcomes.sas` Traj2 (released core) Authors: Weiyi Xia, Haiqun Lin, Anum Zafar

What it does

This file fits group-based trajectory models for one or two continuous outcomes with bounded support, using a censored-normal (Tobit) model in PROC NLMIXED. It sorts subjects into a few latent classes, each with its own polynomial trend over time. Two outcomes can be fit jointly under one shared class structure. Everything is macro-only with no compiled code, so it runs as-is in the CMS VRDC and any SAS 9.4 or newer session.

What you need

- SAS 9.4 or newer with SAS/STAT (PROC NLMIXED).
- ODS Graphics and PROC IML for the plots (`%plot_prep`). The fit itself runs without IML.
- Write access to the WORK library.

The model in brief

Each class j has its own polynomial mean over time, and observations are censored normal: floored at 0 and capped at the per-time value in max_values. For residual e = Y - mu and residual SD sigma, the log contribution at each time point is:

```
interior (0 < Y < cap):  logpdf('NORMAL', e/sigma) - log(sigma)
lower censored (Y = 0):   logcdf('NORMAL', e/sigma)
upper censored (Y = cap): logcdf('NORMAL', -e/sigma)
```

Class membership is a softmax on class intercepts, with class A fixed to 0. The joint model multiplies the two outcome contributions within each class before mixing, which assumes the two outcomes are independent given class. Missing time points are allowed under missing-at-random.

Quick start (simulated data)

Set the values at the top of the file (Step 0) and source it. That single action simulates data, fits both single-outcome models, builds the warm start, fits the joint model, and draws the plots.

```
%let USE_SIM     = 1;    /* 1 = simulate, 0 = your own data */
%let T           = 12;   /* time points */
%let class       = 3;    /* number of classes */
%let order_model = 2;    /* 1 linear, 2 quadratic, 3 cubic */
%let equal       = T;    /* T = one sigma across classes */
%let max_values  = 10 10 10 10 10 10 10 10 10 10 10 10;

%include "/path/to/traj2/continuous_2outcomes.sas";
```

The settings you edit

Setting	Meaning
USE_SIM	1 to simulate test data, 0 to use your own BASE_FILE_SRS.

Setting	Meaning
`T`	Number of time points.
`class`	Number of latent classes (labels A, B, C, ... in order).
`order_model`	Trend shape: 1 linear, 2 quadratic, 3 cubic.
`equal`	T for one shared sigma, F for one sigma per class.
`max_values`	Upper cap per time point (one number each, length T).
`y1vars` / `y2vars`	Outcome columns (defaults `QHH1-QHH12`, `QINP1-QINP12`).
`idvar` / `tvars`	Subject id and time-index columns (defaults `BENE_ID`, `quar1-quar12`).

Using your own data

Set USE_SIM=0 and have a wide table named BASE_FILE_SRS in your session before you source the file, one row per subject, with the columns named in your settings.

Column	What to supply
Subject id	Named by `idvar`, unique per subject.
Outcome 1	Named by `y1vars`, one column per time point.
Outcome 2	Named by `y2vars`. Needed only for the joint fit.
Time index	`tvars` (quar1-quarT) holding the values 1..T. Create these yourself.

Two things to get right:

- Floor at 0. The lower censoring point is fixed at 0, so rescale if your outcome floor is not 0.
- Caps must match reality. Check max_values against each outcome's sample maximum, or cases at the cap will be treated as interior.

Running the pipeline

The run block at the bottom of the file does four things in order: fit outcome 1, fit outcome 2, build the warm start, then fit the joint model and plot. You can also call these yourself.

Single-outcome fit

```
%nlmixed_1(T=&T, LC=&class, Y=1,
  starting=%starting_value_alpha(class=&class)
           %starting_value_beta_sigma(class=&class, outcome=1,
             order=&order_model, equal_sigma=&equal),
  output=nlm_fix_T1&class, order=&order_model, equal_sigma=&equal);
```

Warm start, then joint fit

The joint fit is sensitive to starting values, so it is seeded from the two single fits: stack them, drop the alpha rows, and pass the result as starting values. Always run the single fits first.

```
data work.nlm_2y_starting;
  set nlm_fix_T1&class nlm_fix_T2&class;
  if parameter =: 'alpha' then delete;
run;

%nlmixed_MultiTraj(T=&T, LC=&class,
  starting=%starting_value_alpha(class=&class) / data=work.nlm_2y_starting,
  output=nlm_fix_T1_T2&class, order=&order_model, equal_sigma=&equal);
```

What you get back

- `nlm_fix_T1{class}`, `nlm_fix_T2{class}`: the two single-outcome fits.
- `nlm_fix_T1_T2{class}`: the joint fit, which is the main result.
- `avg_membership`: average class sizes, plus predicted and predicted-vs-observed plots from `%plot_prep`.

Fit statistics (AIC, BIC, and so on) print in the NLMIXED output. To choose the number of classes, refit with different class values and prefer the lowest BIC with sensible class sizes.

Troubleshooting

Symptom	Fix
Joint fit will not converge	Run the single fits first so the warm start exists. If it still struggles, try fewer classes or a lower order_model.
Predicted curves look wrong	Check max_values against the sample max, and pass the same equal_sigma to %plot_prep that you fit with.
Lower censoring never fires	The outcome floor is not 0. Rescale so the minimum is 0.
Time-index error with USE_SIM=0	quar1-quarT were not filled with 1..T. Create them before sourcing the file.